

TASK 15: Multithreading – Ticket Booking Simulation

Tools:

- Primary: IntelliJ IDEA, Java Threads
- Alternatives: Eclipse

Hints / Mini Guide:

1. Create multiple threads representing users booking tickets.
2. Use shared resource representing ticket count.
3. Demonstrate race condition.
4. Fix concurrency issues using synchronized.
5. Implement thread-safe booking logic.
6. Use sleep() to simulate real delay.
7. Explain thread lifecycle.
8. Print booking logs clearly.

Deliverables:

- Multithreaded booking simulation
- Output showing synchronized execution

Interview Questions Related To Above Task:

- What is multithreading?
- What is race condition?
- synchronized vs Lock?
- Thread lifecycle?
- Deadlock meaning?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

